DAVE V. DELA CERNA

I. TRUE OR FALSE. Write **TRUE** if the statement is True and **FALSE** if it is incorrect.

FALSE 1. Antivirus alone is enough to secure an operating system.

FALSE 2. Network segmentation reduces the spread of cyberattacks. (TRUE)

TRUE 3. SaaS users are responsible for securing physical servers. (TRUE)

FALSE 4. Containers share the host OS kernel.

TRUE 5. Strong passwords eliminate all cyber threats.

TRUE 6. Firewalls can filter both incoming and outgoing traffic.

FALSE 7. Cloud security is solely the provider's responsibility.

TRUE 8. Virtual machines are more isolated than containers.

FALSE 9. Zero Trust means trusting all internal users.

FALSE 10. Emerging technologies always eliminate security risks.

II. Scenario-Based. Understand each scenario and answer the question that follows. (2 pts. each)

1. A company failed to update its OS and got hacked. What security practice was ignored?
   -Regular Operating System Updates

2. Employees can access all files in a system. What principle is violated?
   -Principle of the lease privileges

3. A network is divided into multiple segments. What is the purpose of this design?
   -So that it can limit the spread of attacks on all the unit that my business have.

4. A business uses cloud services but data is leaked due to weak passwords. Who is responsible?
   -The user or clients, also the business because they didn't properly instruct the user/client to use strong password and enable MFA.

5. A hacker spreads malware across all computers quickly. What could have prevented this?
   -It must that the user use network segmentation or strong firewalls.

6. A company uses containers but experiences data leaks. What security measure might be missing?
   -Container security is not that so strong also isolation and access control.

7. A system trusts all internal users automatically. What model should be applied instead?
   -Trust zero model

8. A firewall allows all traffic. What is the risk?
   -Authorization access and anonymous attacks.

9. A company uses outdated security tools. Why is this dangerous?
   -Because, so hard or to slow to detect and new threats or risk to secure the system.

10. AI detects unusual login attempts. What type of security approach is this?
    -AI automation detect.

III. Performance Task. (5 pts. each)

1. Propose a secure network design using segmentation for a small office.
   -First of all, make the system had firewalls to avoid malicious attacks, and then put the network segmentation on all the unit that the office had. Also, always regular update on the units and the systems.

2. Explain how you would secure data in a cloud environment.
   - Most use strong password and enable multi-factor authentication, encrypting data, and regularly update security settings.

3. Compare virtualization and container security. Which is more secure and why?
   - Virtualization is more secured because each virtual machine has its own full OS, making it isolated. While the containers share the same Operating system, so if one is attacked, others may be affected easily.

4. Discuss one emerging cybersecurity trend and how it helps organizations.
   -The trend nowadays people are using Ai tool so why not to use AI? That detect threats to helps business by quickly detecting unusual activities and stopping attacks faster than ever that can human can do.